company to company, and perhaps over time or with other variables within the same company.

What we can do, though, is providing our containers with a way to filter points in a *NN* search by some conditions that we can pass in our code, thus allowing who uses the container to reason about the business rules, customize them, and pass them as arguments to the search method.

11.2.1 Filtering Points

This way we provide a *template* method that can be effectively customized depending on the business needs.

Listing 11.4 shows the code for k-d tree's nearest neighbor search, modified to allow filtering of points.

As you can see, we pass an extra argument (with respect to the regular method in listing 9.9), a predicate that is evaluated on current node before accepting this point as a nearest neighbor.

The only difference with the basic version, besides boilerplate to pass around this new argument, is at line #5, where the predicate is actually evaluated on current node.

This version of the method allows us to filter single points and solve the first of the issues mentioned in previous section, making sure that we choose a shop that actually has the item(s) a customer ordered.

Listing 11.4 The filteredNearestNeighbor method